

CS 171: Discussion Section 3 (Sept 15)

1. Pseudorandom Generators

Let $F, G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be pseudorandom generators. For each of the functions below, prove or disprove that H is necessarily a pseudorandom generator.

- (a) $H(s_0 s_1 \dots s_{n-1}) := G(s_{n-1} s_{n-2} \dots s_0)$.
- (b) $H(s) := G(s)_{0, \dots, 2n-1}$ (i.e., the first $2n$ bits of $G(s)$).
- (c) $H(s) = G(s) \| F(s)$.

2. Unpredictability and Pseudorandomness

Consider an efficiently computable keyed function $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$. We'll define the following game $\text{Pred}_{\mathcal{A}, F}(n)$:

1. Sample $k \leftarrow \{0, 1\}^n$ uniformly and set $O := F_k(\cdot)$.
2. $\mathcal{A}$ is given 1^n and oracle access to $O(\cdot)$: it may query any $x \in \{0, 1\}^n$ and receives $O(x)$. Let Q be the set of inputs queried by $\mathcal{A}$.
3. $\mathcal{A}$ outputs a prediction $(x^*, y^*) \in \{0, 1\}^n \times \{0, 1\}^n$.
4. Output 1 if $x^* \notin Q$ and $y^* = O(x^*)$, and 0 otherwise.

We say that F is *unpredictable* if for every probabilistic polynomial-time adversary $\mathcal{A}$:

$$\Pr[\text{Pred}_{\mathcal{A}, F}(n) = 1] \leq \text{negl}(n).$$

- (a) Why does the definition require that x^* was never queried?
- (b) Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a secure PRF. Prove that F is unpredictable.
- (c) Does unpredictability imply pseudorandomness? Either prove the implication or construct a function family F that is unpredictable but not pseudorandom, and prove both properties.